

SUPERMARKET SALES ANALYSIS

Professional Data Analysis Report

Executive Summary: The dataset contains 9,994 transaction rows covering January 2014 to December 2017. Total sales were \$2,297,200.86, total profit was \$286,397.02, and the average order value was \$458.61. The analysis identifies category, regional, temporal and discount-related patterns while preserving loss-making transactions for investigation.

1. Dataset & Data Quality

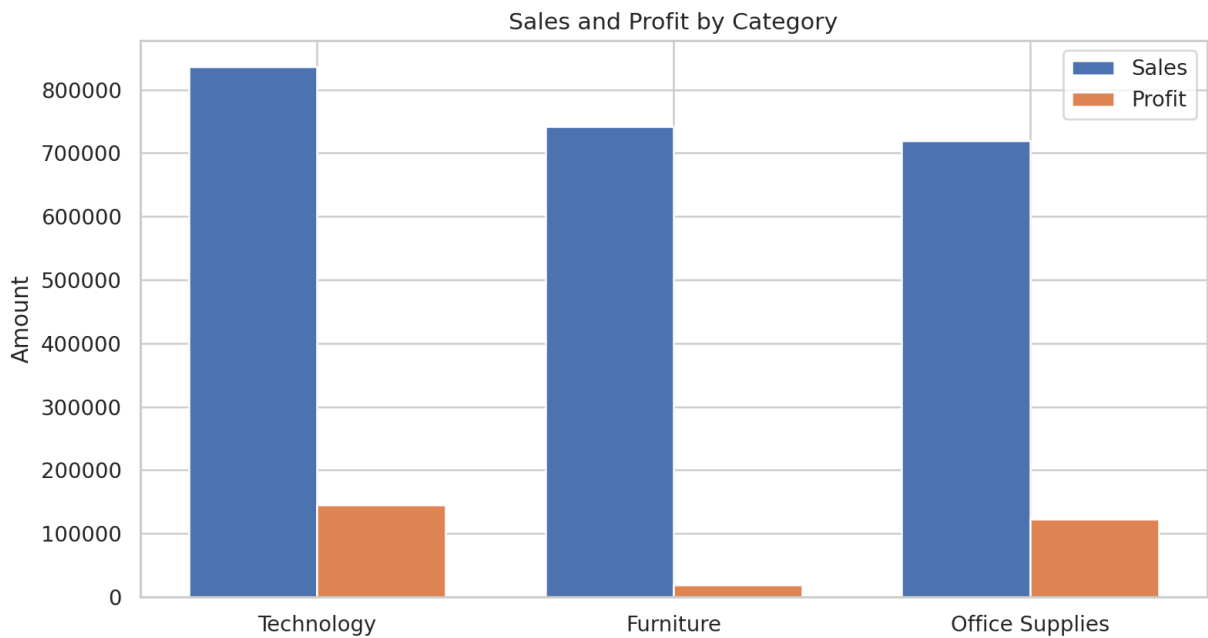
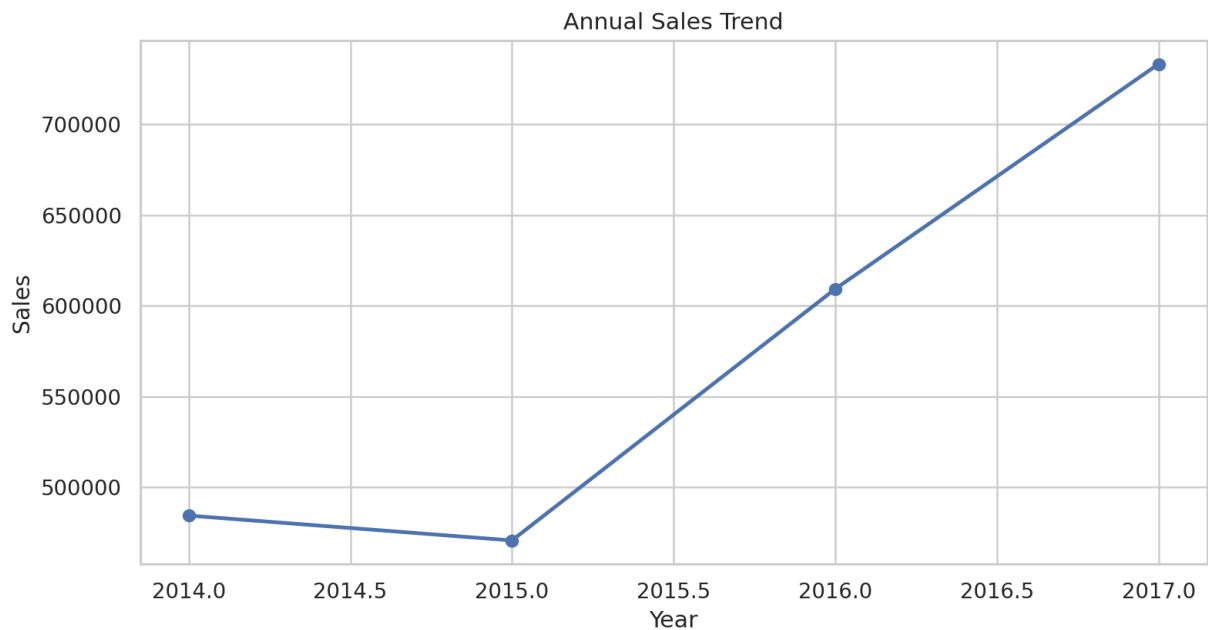
The dataset contains customer, order, product, geographic, sales, quantity, discount and profit fields. There are no missing cells and no duplicate rows. Dates were converted to datetime and derived fields were created for year, month, weekday and shipping duration.

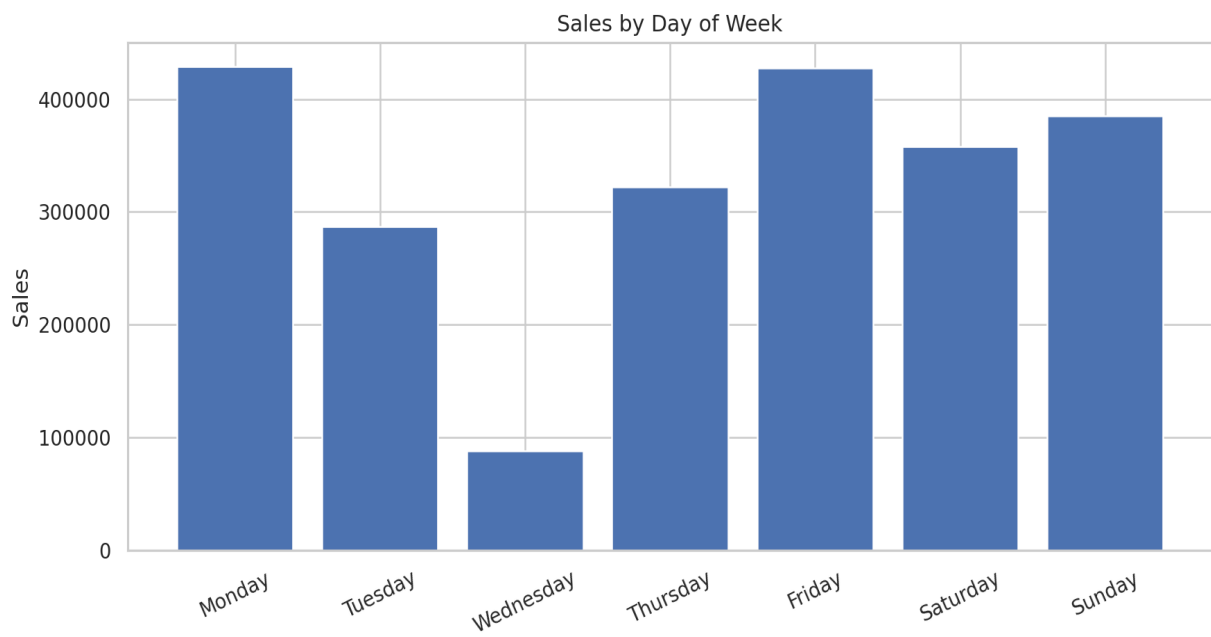
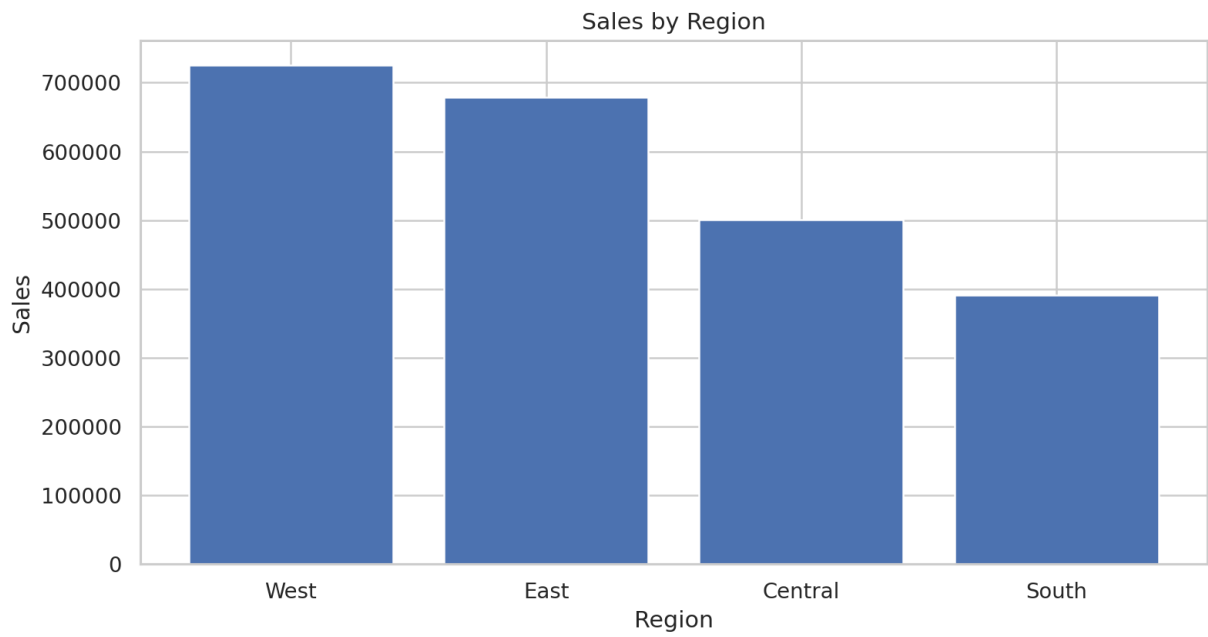
Metric	Value
Rows	9,994
Unique Orders	5,009
Unique Customers	793
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Quantity	37,873
Average Order Value	\$458.61
Profit Margin	12.47%

2. Key Findings

- Technology is the highest-sales category at \$836,154, while Office Supplies has the highest quantity sold (22,906 units).
- The West region records the highest sales at \$725,458.
- Annual sales increased from \$484,247 in 2014 to \$733,215 in 2017.
- Tables have high sales (\$206,966) but negative aggregate profit (\$-17,725), indicating a margin-management issue worth investigating.
- There are 1,871 loss-making transaction rows. This makes profit-focused analysis important rather than relying on sales alone.
- Discount and profit have a correlation of -0.219; this is an association and does not by itself prove that discounts cause lower profit.

3. Visual Analysis





4. Recommendations

- Review pricing, discounting and product-level economics for loss-making sub-categories, particularly Tables and Bookcases.
- Study the operating and product mix factors behind strong West-region sales before attempting to replicate them in other regions.
- Track gross profit and margin alongside revenue when evaluating products and promotions.
- Use customer segment and product-level discount analysis before changing promotion rules.
- Automate this analysis on a recurring schedule so management can monitor changes in sales, profit and loss-making products.

5. Limitations

The dataset does not include inventory, acquisition cost, marketing spend or customer satisfaction. Therefore, the analysis can identify patterns and associations but cannot establish causal effects or fully explain profitability. Findings are specific to the supplied dataset and its 2014–2017 period.

STUDENT PERFORMANCE ANALYSIS

Portfolio Data Analysis Report — Simulated Dataset

Executive Summary: This project analyzes a simulated dataset of 600 students. The analysis examines pass/fail outcomes, subject averages, attendance, study hours and their relationships with final performance. The overall simulated pass rate is 100.00%. Attendance and study hours are positively associated with final score in this generated dataset.

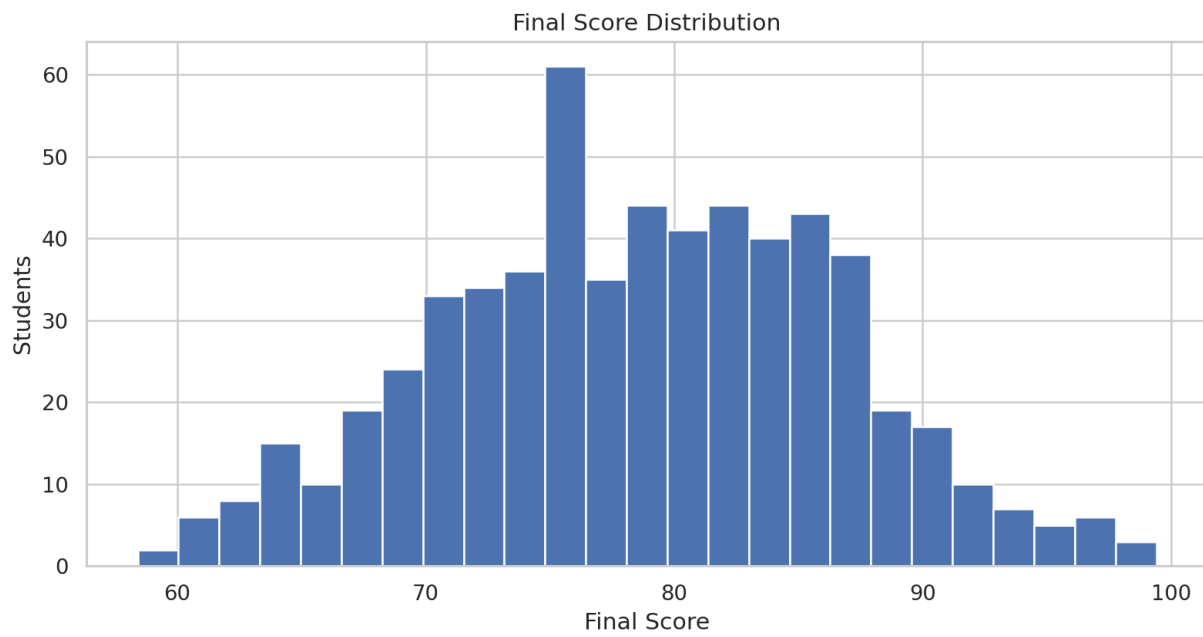
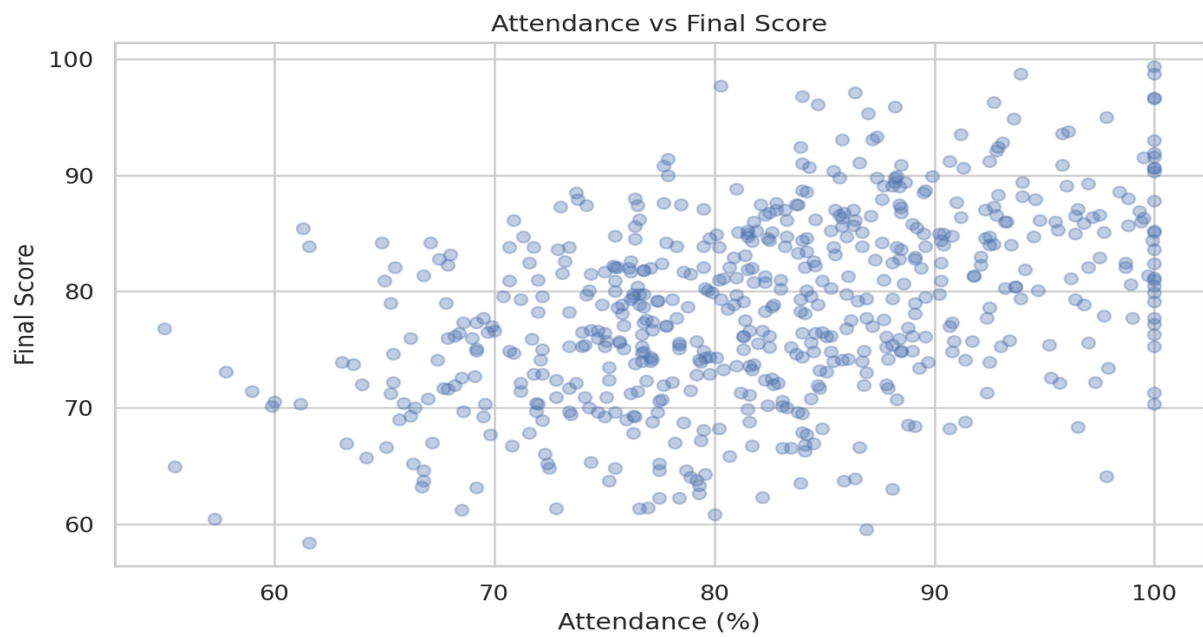
1. Data Quality

The dataset contains 600 rows and 10 columns. Validation confirms no missing cells, no duplicate rows, and valid 0–100 ranges for attendance and scores.

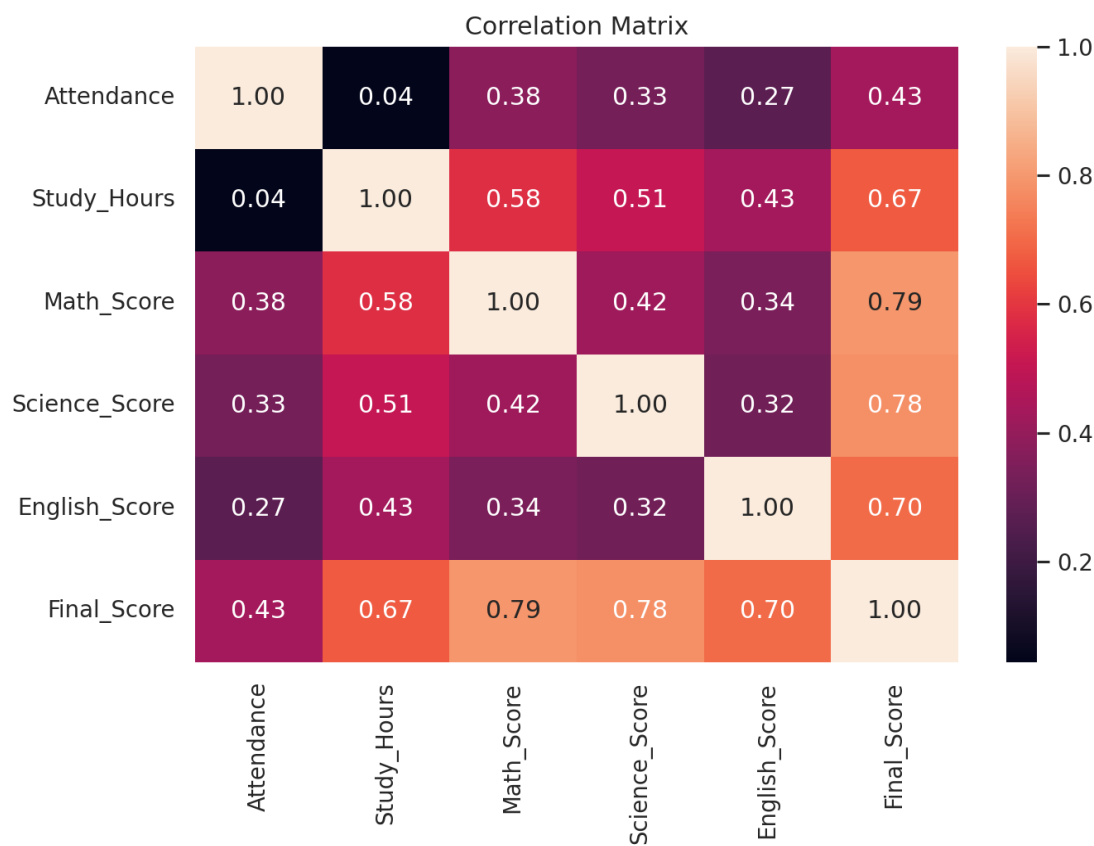
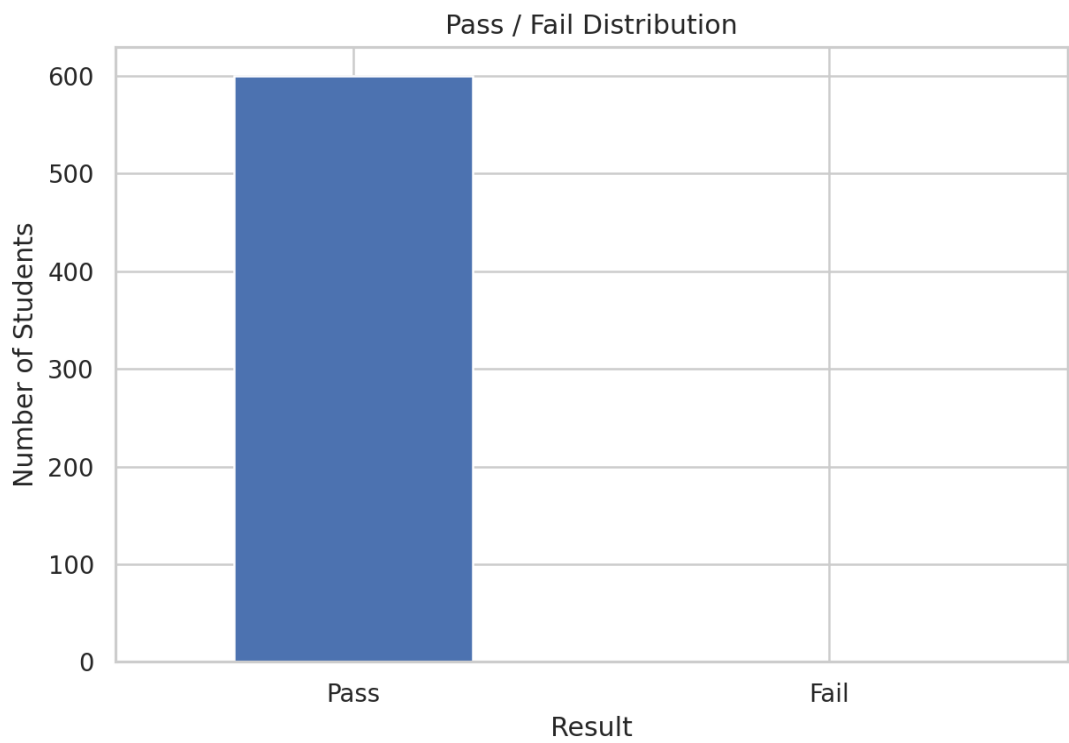
Metric	Value
Students	600
Pass rate	100.00%
Fail rate	0.00%
Attendance / Final correlation	0.433
Study hours / Final correlation	0.671
High attendance mean	82.53
Low attendance mean	72.91
Welch t-test p-value	5.85176e-18

2. Visual Analysis





3. Additional Analysis



4. Key Findings

- The simulated pass rate is 100.00%.
- Attendance and final score correlation is 0.433.
- Study hours and final score correlation is 0.671.
- Students with attendance $\geq 85\%$ average 82.53, compared with 72.91 for students below 70% attendance.
- The Welch t-test p-value is 5.85176e-18; interpretation should use a preselected significance level and recognize that the data are simulated.

5. Recommendations

- Use attendance as one component of an early-warning framework.
- Provide targeted academic support to students showing both low attendance and low scores.
- Compare subject-level outcomes to identify areas for additional support.
- Monitor study habits together with academic outcomes.
- Validate findings with real institutional data before making operational decisions.

6. Limitation

This dataset is simulated and is intended only for demonstrating data-analysis techniques. The findings must not be represented as evidence about a real student population.

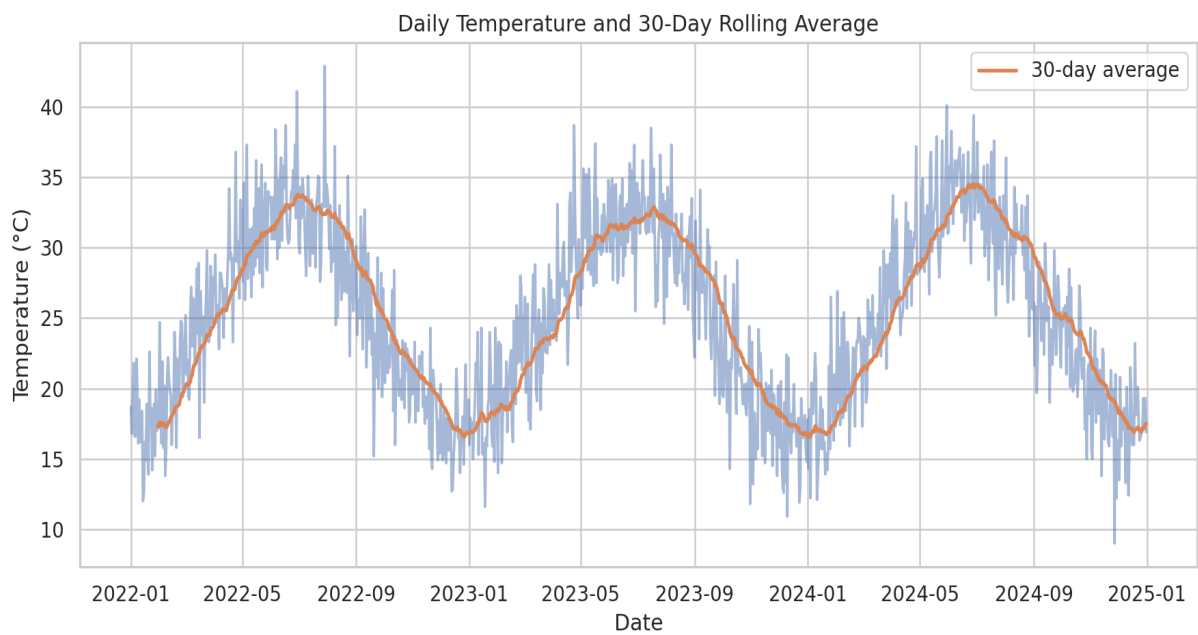
WEATHER DATA ANALYSIS

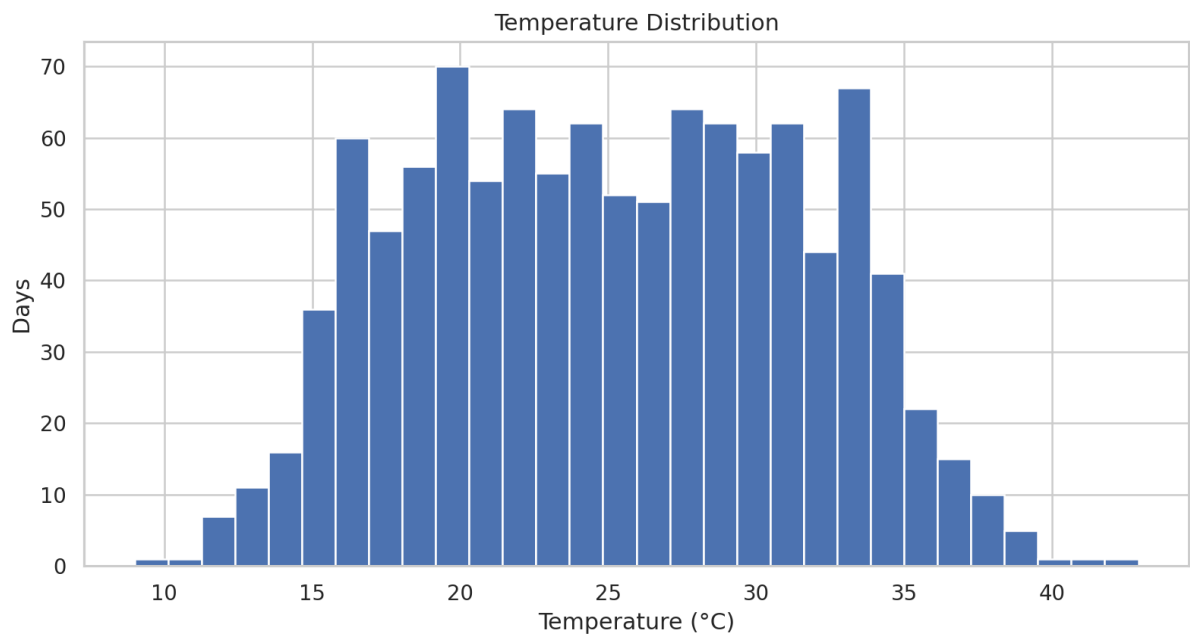
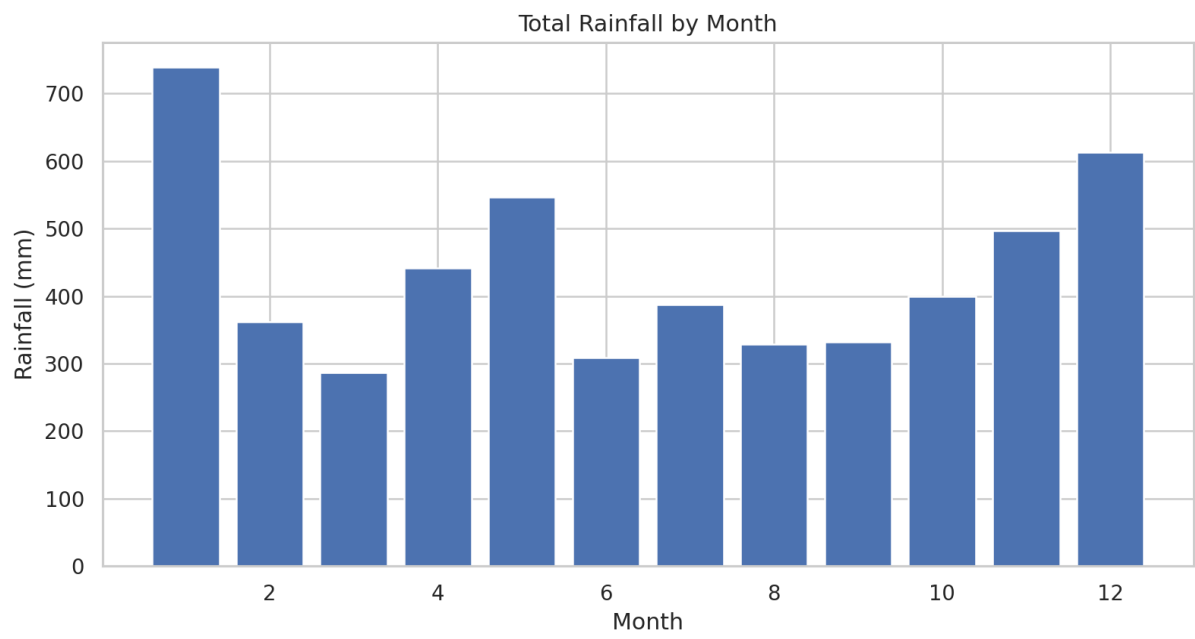
Portfolio Data Analysis Report — Simulated Dataset

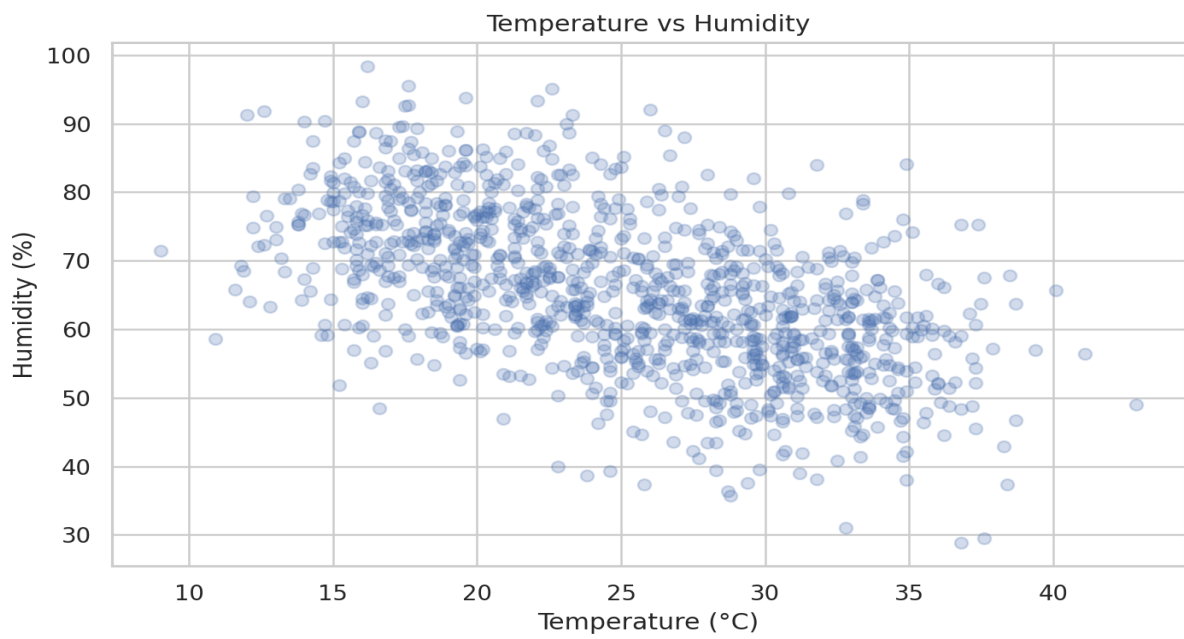
Executive Summary: This project analyzes 1,096 simulated daily observations from 2022-01-01 through 2024-12-31. The analysis focuses on temperature trends, seasonal rainfall, humidity relationships and extreme observations.

Metric	Value
Observations	1,096
Period	2022-01-01 to 2024-12-31
Average temperature	25.10 °C
Total rainfall	5,240.8 mm
Rainy days	279
Temp / humidity correlation	-0.565
Temp / rainfall correlation	-0.080
Extreme-temperature days	41

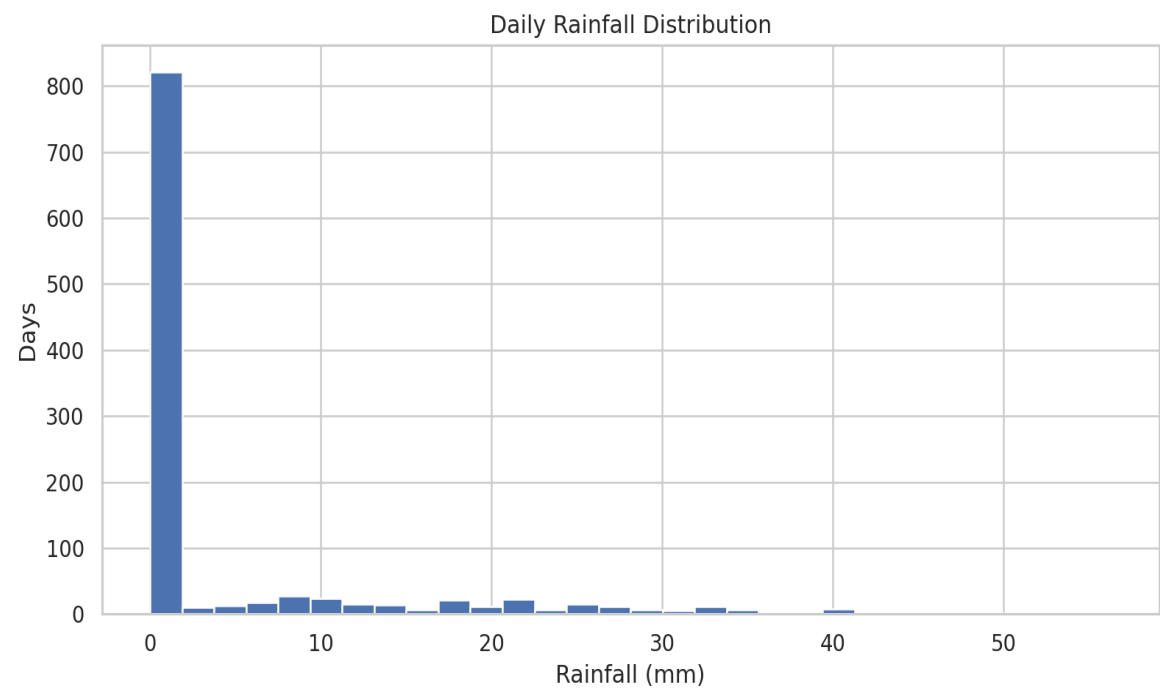
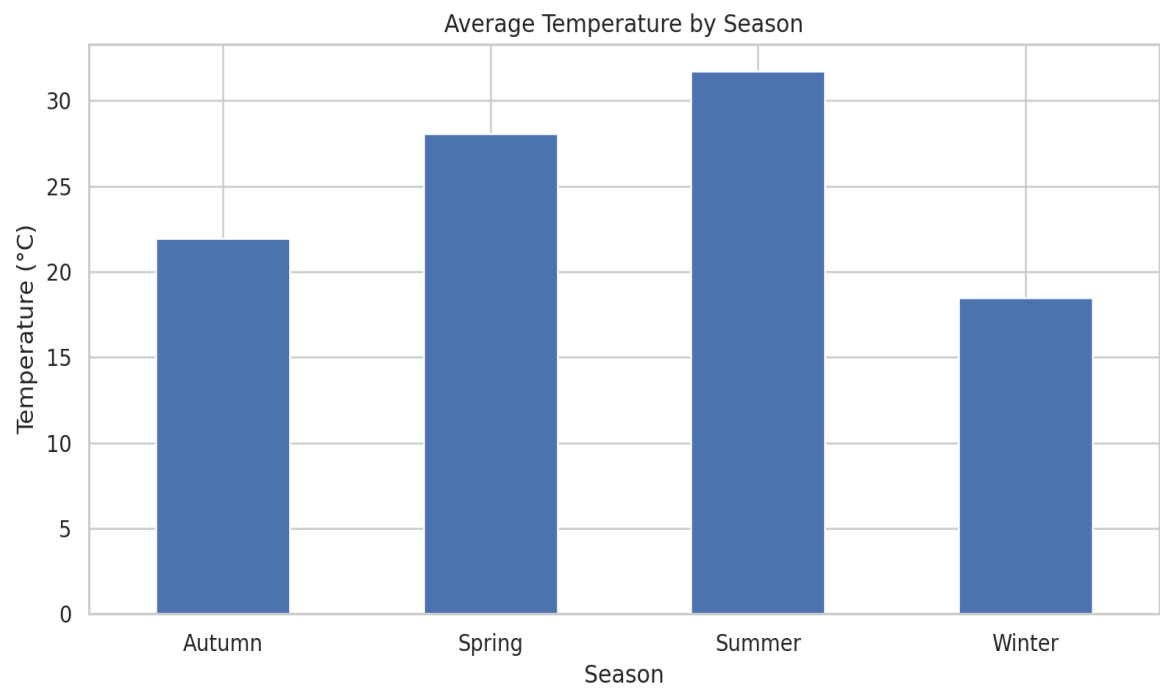
1. Visual Analysis







2. Seasonal Analysis



3. Key Findings

- Temperature exhibits a recurring seasonal pattern.
- The strongest monthly rainfall total occurs in month 1.
- The simulated temperature/humidity correlation is -0.565.
- The simulated temperature/rainfall correlation is -0.080.
- The hottest recorded observation is 42.9 °C on 2022-07-29.
- The wettest recorded observation is 56.3 mm on 2022-03-06.

4. Recommendations

- Use monthly and seasonal summaries for planning.
- Monitor rainfall extremes separately from average rainfall.
- Use rolling averages to distinguish sustained trends from daily noise.
- Use validated, longer-term observational data for real meteorological or climate decisions.

5. Limitation

The dataset is simulated and is intended for demonstrating analysis techniques rather than representing official weather observations.

Project 4 — Healthcare Patient Analysis

Dataset note: All records are simulated for portfolio practice and are not suitable for clinical use.

Executive Summary

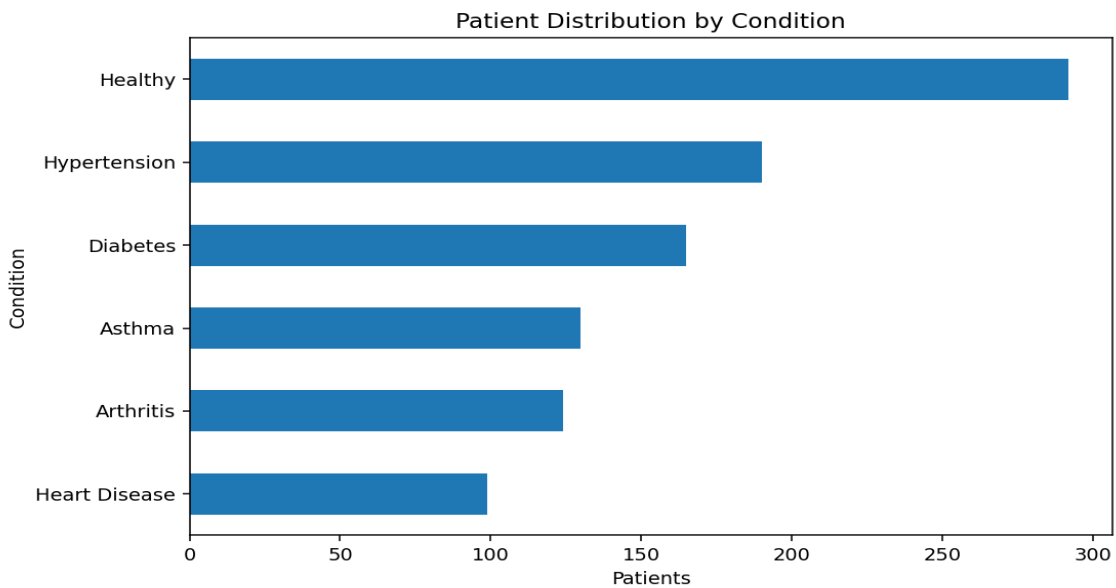
The cleaned dataset contains 1,000 synthetic patient records. Average age is 51.7 years, average BMI is 26.7, and average annual visits are 4.35. The synthetic high-risk label occurs in 30.2% of records.

Data Quality

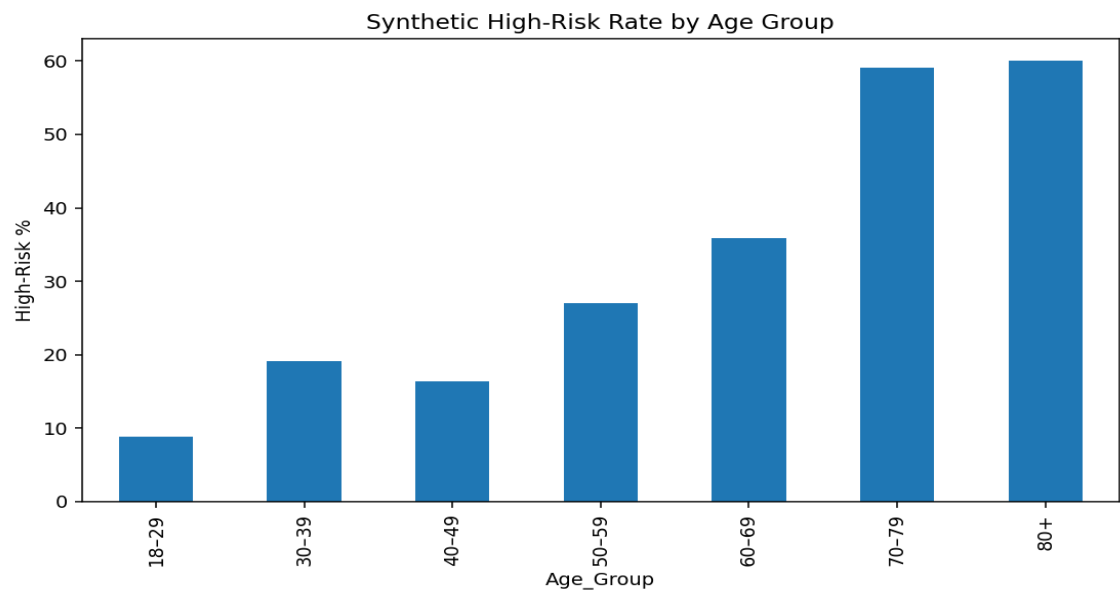
Metric	Result
Raw rows	1,002
Clean rows	1,000
Duplicates removed	2
Missing numeric fields	Median imputation

Condition Summary

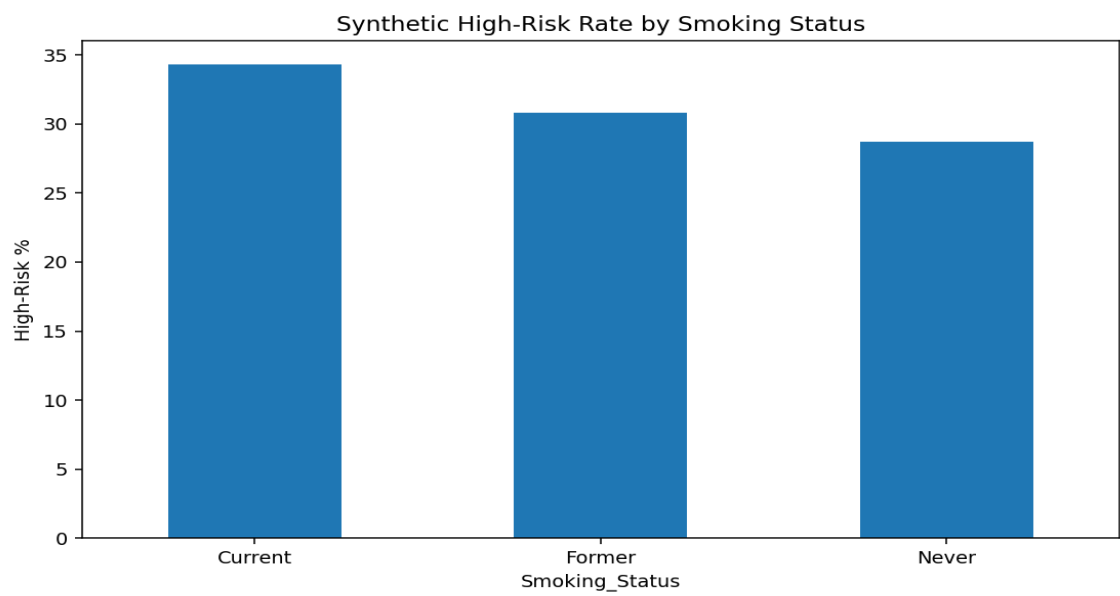
Condition	Patients	Avg Age	Avg Visits	Risk %
Healthy	292	51.3	3.80	26.4%
Hypertension	190	51.4	4.54	28.9%
Diabetes	165	53.7	4.63	39.4%
Asthma	130	51.3	4.61	23.8%
Arthritis	124	52.5	4.62	34.7%
Heart Disease	99	49.7	4.43	31.3%



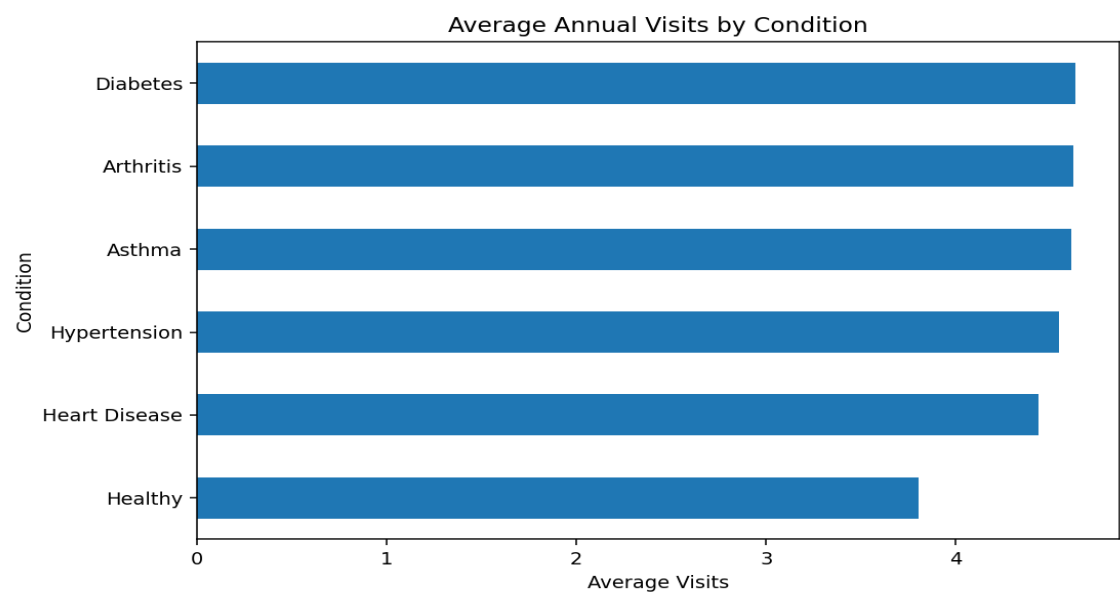
Patient distribution by condition



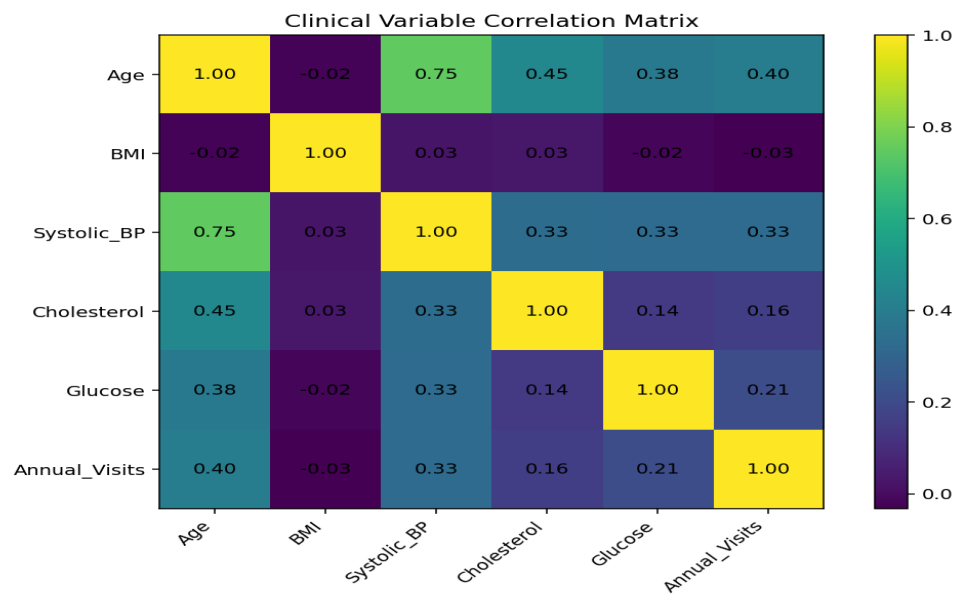
Synthetic high-risk rate by age group



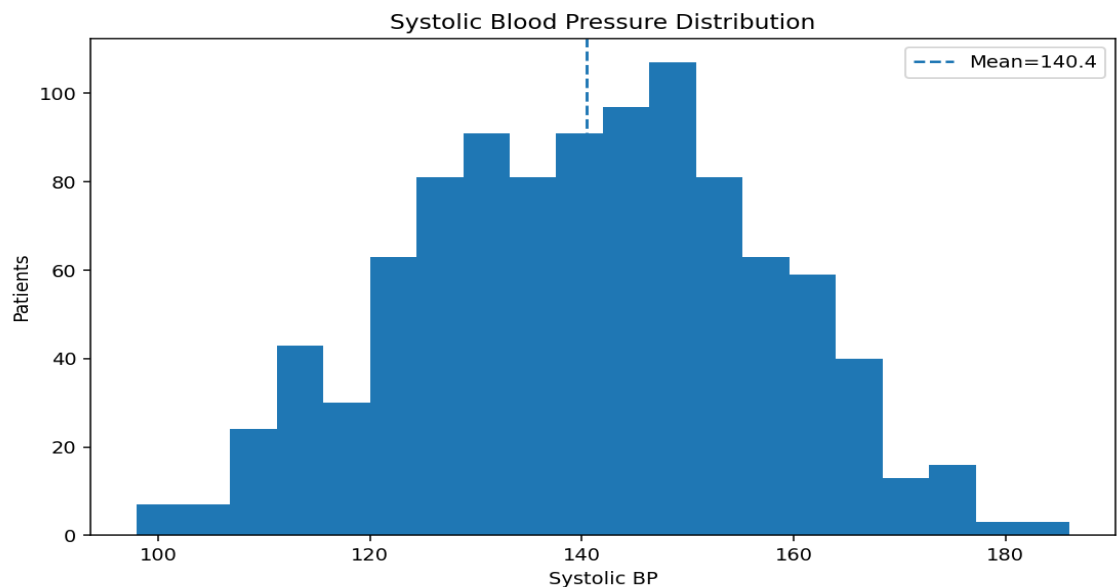
Synthetic high-risk rate by smoking status



Average annual visits by condition



Clinical-variable correlation matrix



Systolic blood pressure distribution

Insights and Responsible Interpretation

Condition, age, smoking status and visit patterns provide useful examples of healthcare segmentation. Correlations can identify relationships for further investigation but do not prove causation. Because every observation is synthetic, none of these results should be interpreted as medical evidence or generalized to real patients.

Project 5 — Finance Market Analysis

Dataset note: This is simulated financial data for portfolio practice. It is not real market data or investment advice.

Executive Summary

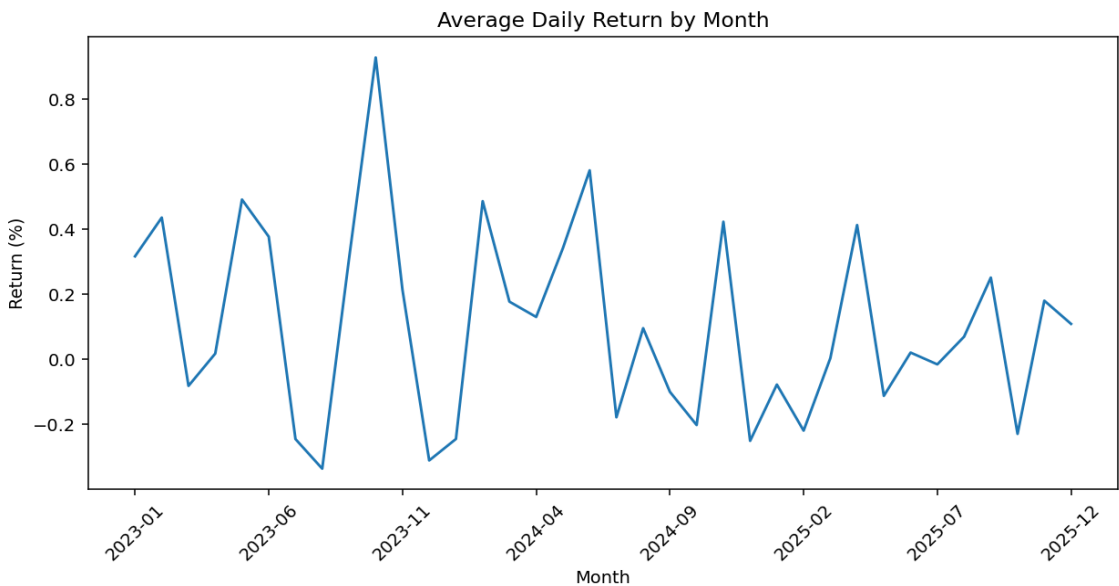
The cleaned dataset contains 1,500 synthetic observations. Average daily return is 0.099% and estimated annualized volatility is 28.13%. 53.2% of observations have positive daily returns. Sector and sentiment analyses demonstrate descriptive financial analytics while avoiding predictive claims.

Data Quality

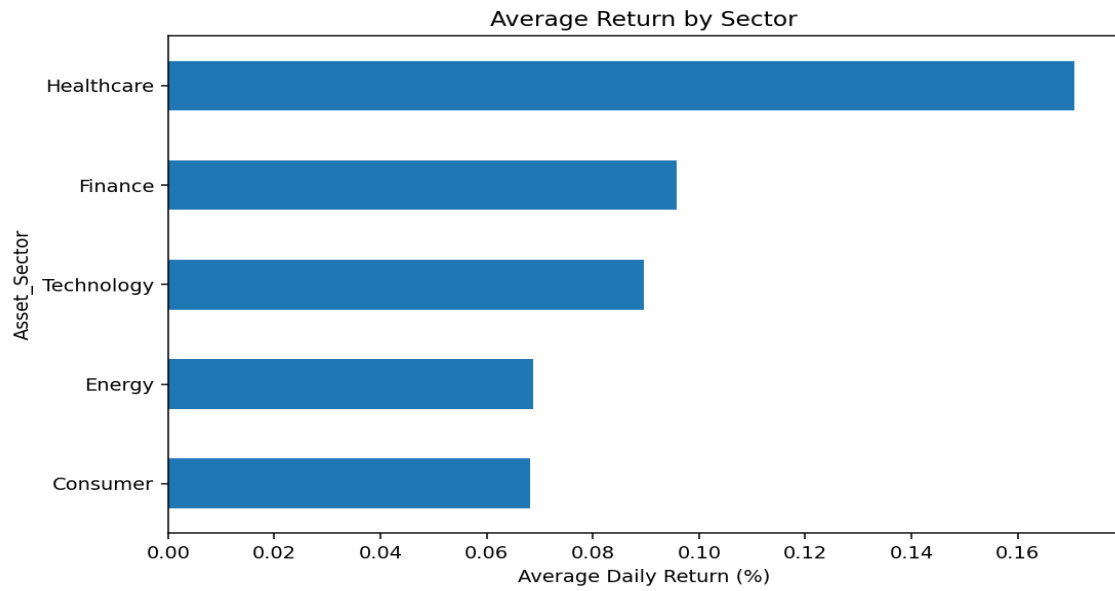
Metric	Result
Raw rows	1,502
Clean rows	1,500
Duplicates removed	2
Missing values	Median imputation

Sector Summary

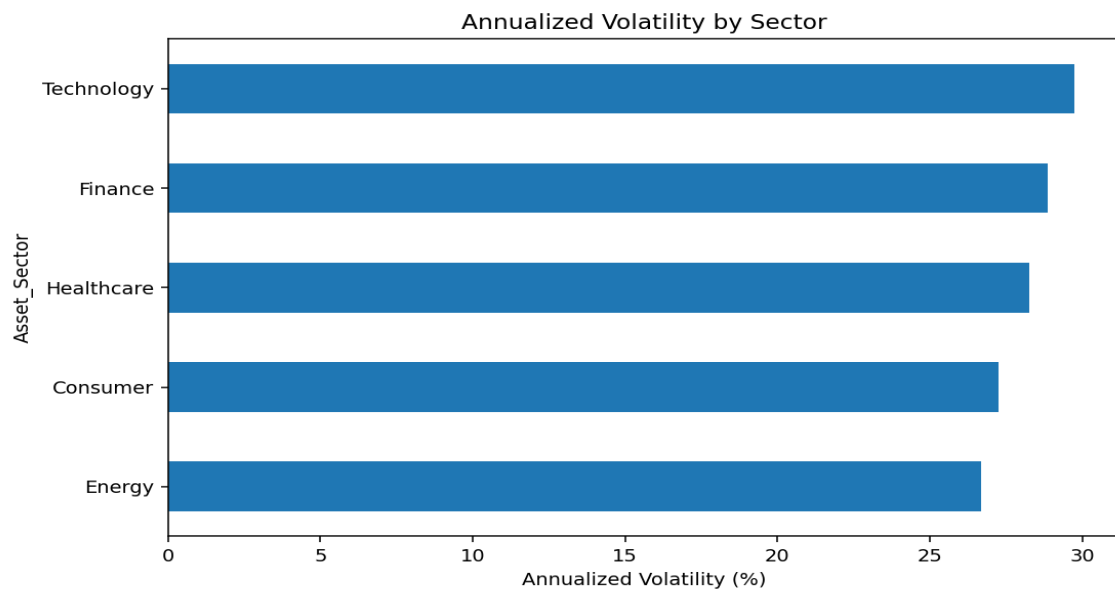
Sector	Avg Return %	Volatility %	Avg Volume
Healthcare	0.171	28.27	565,594
Finance	0.096	28.88	559,331
Technology	0.090	29.74	539,272
Energy	0.069	26.68	532,242
Consumer	0.068	27.26	537,527



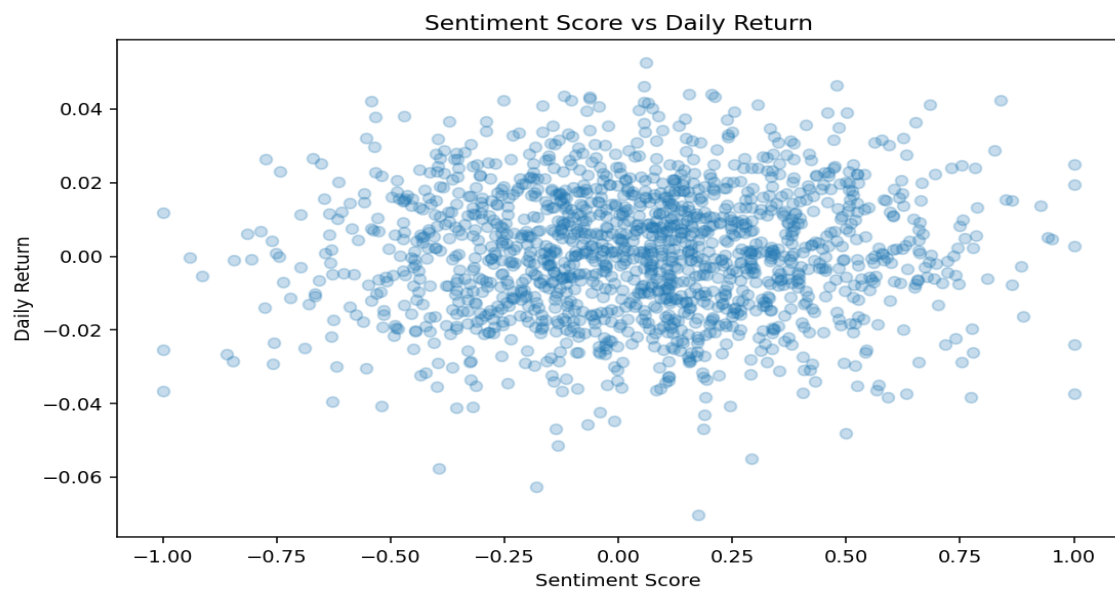
Average daily return by month



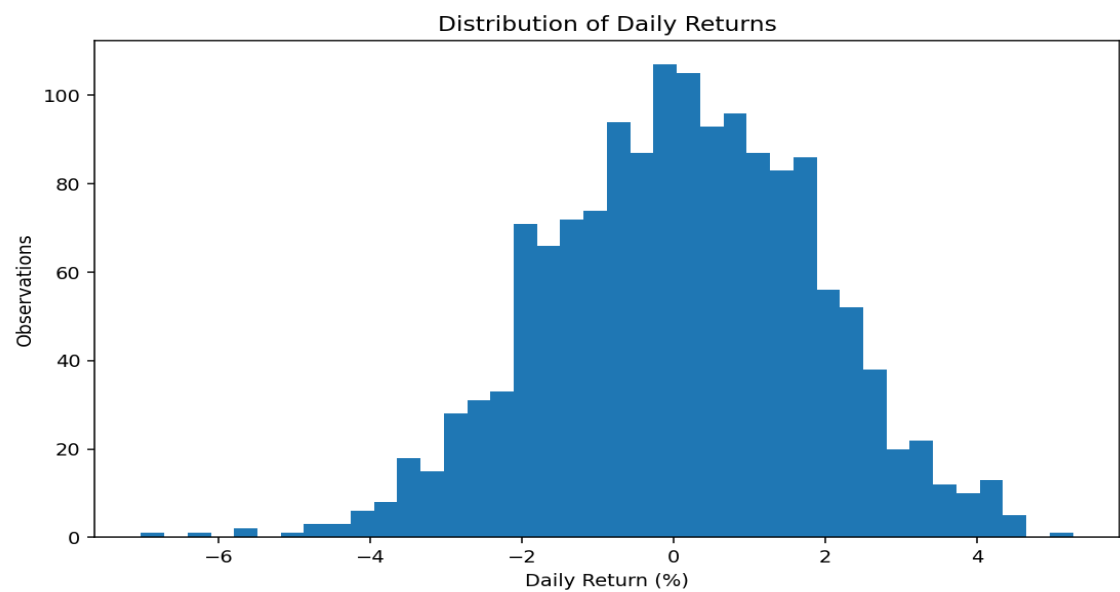
Average return by sector



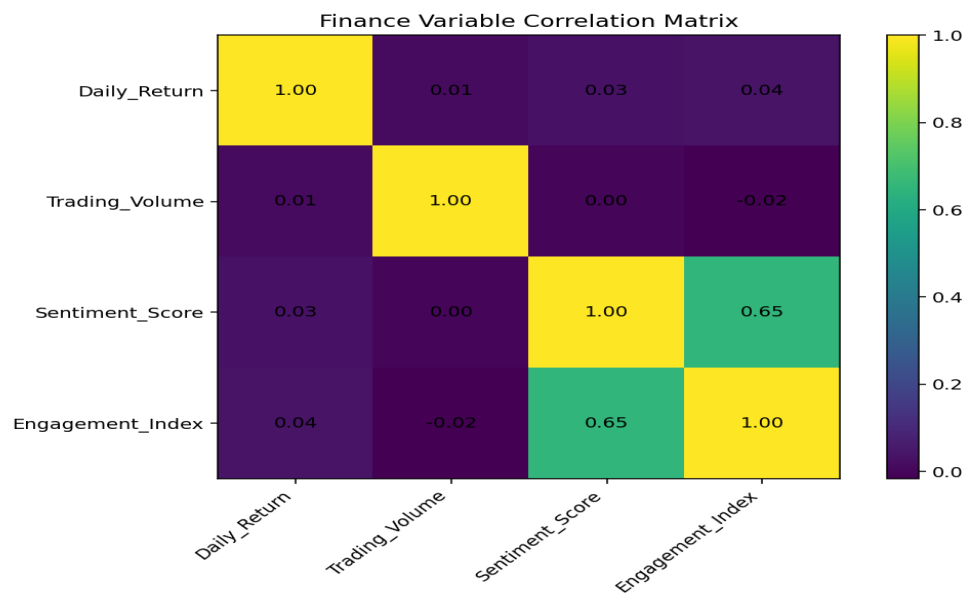
Annualized volatility by sector



Sentiment score versus daily return



Distribution of daily returns



Finance-variable correlation matrix

Interpretation and Limitations

Return and volatility are descriptive measures from simulated observations. Correlation can identify relationships worth investigating but cannot establish causation. The dataset is synthetic, so no result should be generalized to real markets or used to make investment decisions.